



Phase 0 - Part 2

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Q Why do we need virtual environment?

Think of virtualenv like a separate kitchen for each project:

① Project A needs Fastapi == 0.100

② Project B needs Fastapi == 0.95

if both use the same global Python
→ dependency fight.

So we create separate environment.

Function

① Handle request

② Process data

③ Talk to Database

④ Return responses

ex- `def get_user_age(year_of_birth: int) → int:`

`current_year = 2026`

`return current_year - year_of_birth`

① `year_of_birth: int` → Type hint

② `→ int` → Function return int

Why type hint?

① FastAPI uses them for validation

② Better autocomplete

③ Fewer bugs, clean code

Module and Import.

Instead of one big file:

`# utils.py`

`def add(a: int, b: int) → int:`

`return a + b`



```
# main.py
```

```
from Utils import add
```

```
result = add(2,3)
```

```
Print(result)
```

Analogy:

○ Don't keep kitchen, billing, storage in one room

○ Split responsibilities

④ classes and objects

Class UserServices:

```
def __init__(self, users: list):
```

```
    self.users = users.
```

```
def get_user_count(self) -> int:
```

```
    return len(self.users)
```

Usage:

```
Service = UserServices(["A", "B", "C"])
```

```
Print(service.get_user_count())
```

Class = Blueprint of a machine

object = Actual machine built from it.

⑤ Type Hint

```
def create_user(name: str, age: int) -> int:
```

```
    return {"name": name, "age": age}
```

FastAPI use this to:

○ validate input

○ Generate docs

○ Prevent wrong data types



Error Handling

```
def divide (a: int, b: int) -> float:
```

```
    try:
```

```
        return a/b
```

```
    except ZeroDivisionError:
```

```
        return 0.0
```

Analogy:

Instead of restaurant closing
because gas is off,

you show: "Sorry kitchen unavailable
right now".

Async vs Sync

Sync Function:

```
def get-data():
```

```
    return "data"
```

Async Function:

```
import asyncio
```

```
async def get-data():
```

```
    await asyncio.sleep(1)
```

```
    return "data"
```

Analogy:

○ Sync = one waiter handles one
table at a time

○ Async = one waiter can handle
multiple table while waiting.

FastAPI uses async to handle
many user efficiently.